PORTLAND Public Schools

Facility Improvement & Oversight Committee

April 21, 2026



Agenda

- Introduction
- Shared Values and Commitment to Indoor Air Quality
- Air Quality Educational Specifications and Facility Standards
- Air Quality Implementation in Modernization Design
  - Cleveland
  - Ida B. Wells
  - Jefferson
- Air Purification Unit Inventory and Deployment
- PCEF Grant Award and Projects



Shared Values

- **PPS is committed to the health and safety of children and adults in our buildings,** including indoor air quality.

- **PPS employs a comprehensive strategy to improve indoor environmental health,** including building ventilation design standards, public health protocols, and portable air purifiers.

- PPS designs buildings to meet our decarbonization and climate resilience goals in the **Climate Crisis Response, Climate Justice and Sustainable Practices Policy (CCRP)**





Designing for Healthy Indoor Air



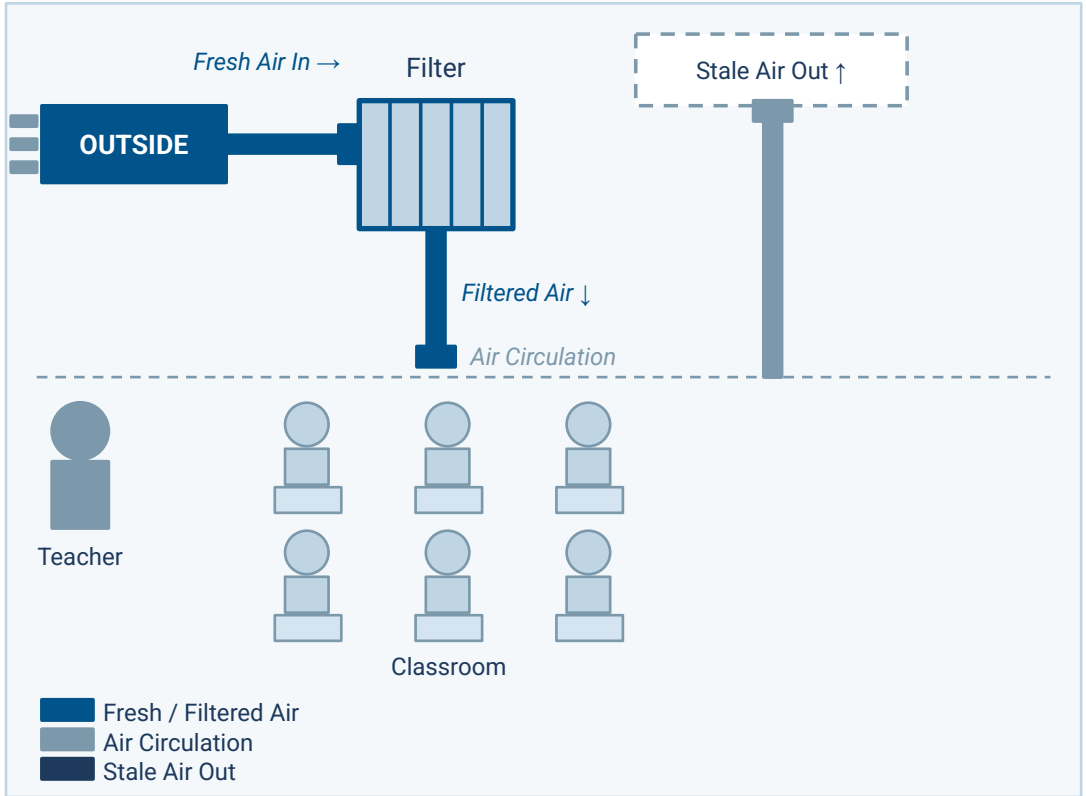




What is Indoor Environmental Quality?

K–12 Indoor Air Quality

Air quality in schools is about ensuring every classroom has clean, comfortable air that supports student learning, health, and focus.
It is managed through how we bring in fresh air, filter it, and maintain safe and comfortable conditions inside our buildings.



KEY COMPONENTS

Ventilation
Fresh outdoor air into classrooms

Filtration
MERV filters remove particles & allergens

Temperature
Comfortable, consistent learning conditions

Humidity
Balanced moisture — not too dry, not too damp

Air Quality Metrics
CO_2 levels show if air is fresh; we also track dust and allergens

WHY IT MATTERS

- Supports student attention and learning
- Improves comfort for students and staff
- Reduces illness and missed school days

Maintained through building systems, regular maintenance, and proactive monitoring.

Goal:

Provide healthy air to enhance academic performance and minimize pathogen transmission.


How Do We Measure
Indoor Air Quality?

QUANTITY

Air Changes Per Hour (ACH)

A measure of air circulation

QUALITY

CO_2 Concentration & Particulates

A measure of air freshness

But ACH is only part of the equation



More circulation or filtration does not necessarily mean better

We need to focus on quality, not quantity



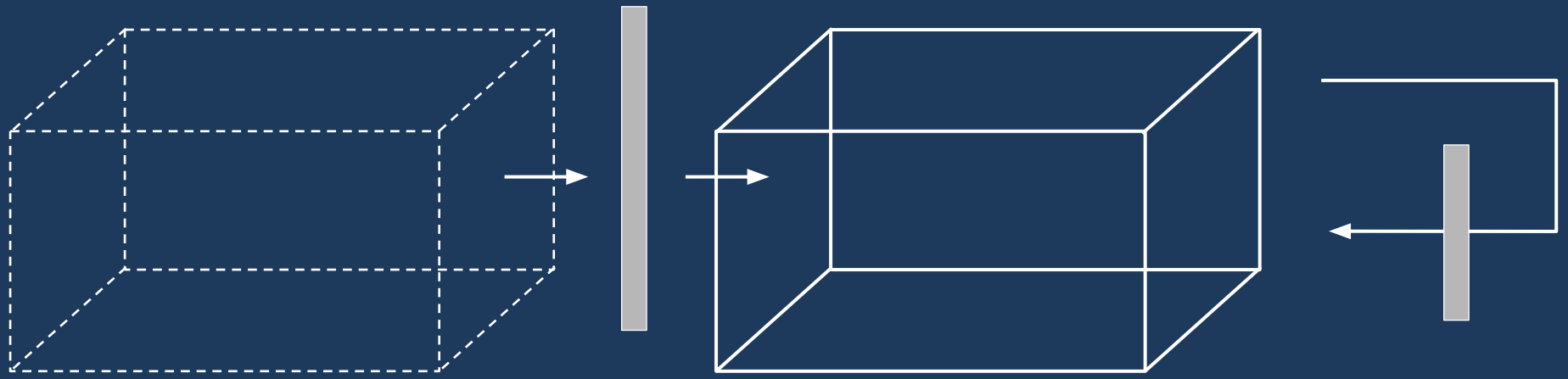


Design to Specifically Address Indoor and Outdoor Pollutants

Addressing Outdoor Pollutants

Outdoor Air In

Recirc Air In



Minimum MERV 13 Filter

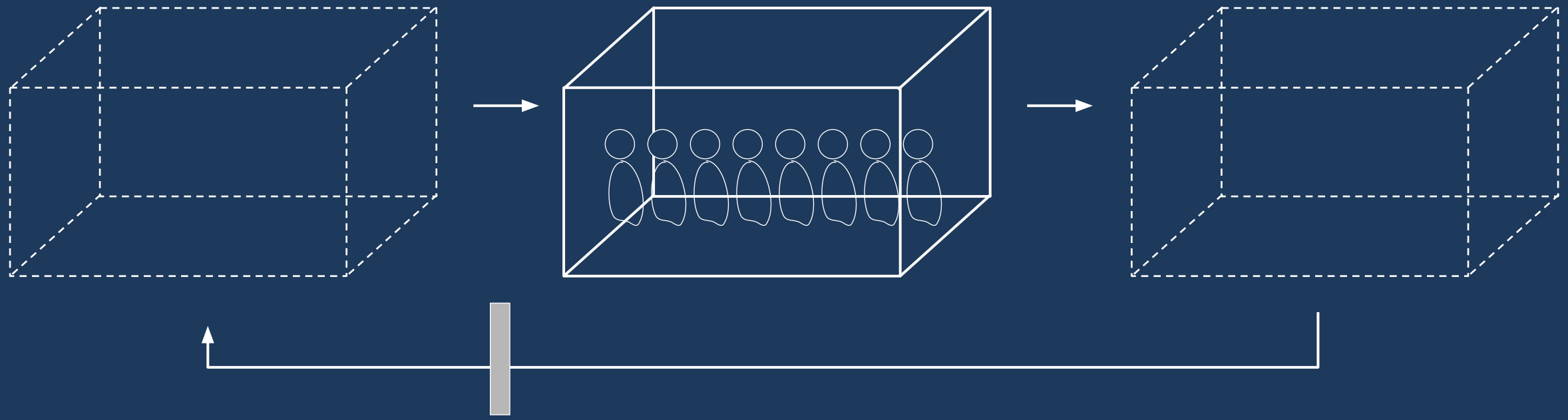
Removes smoke, pollen, and car exhaust

Addressing Indoor Pollutants

Filtered
Outdoor Air In

Generation of CO_2,
Odors, VOCs

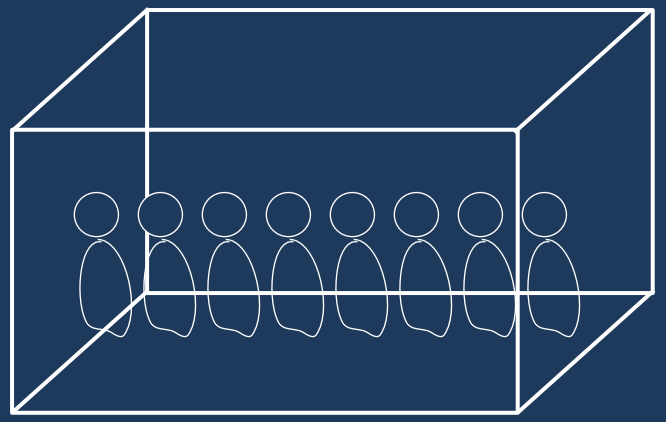
CO_2, Odors,
VOCs Out



recirculating air is filtered with minimum MERV13 as well

Carbon Dioxide Concentration

As a Holistic Air Quality Metric

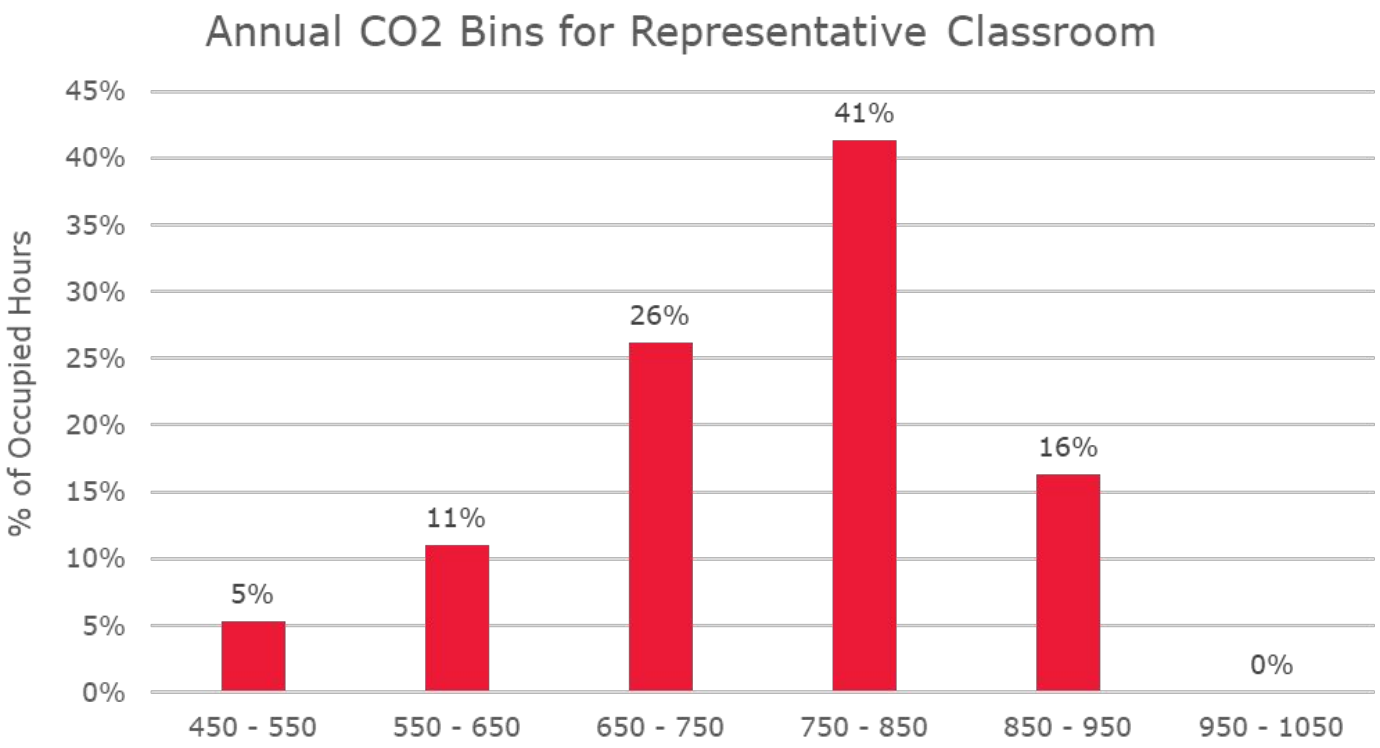


CO_2 should remain ~800 ppm

We are designing for this value.

PPS IAQ Design Summary

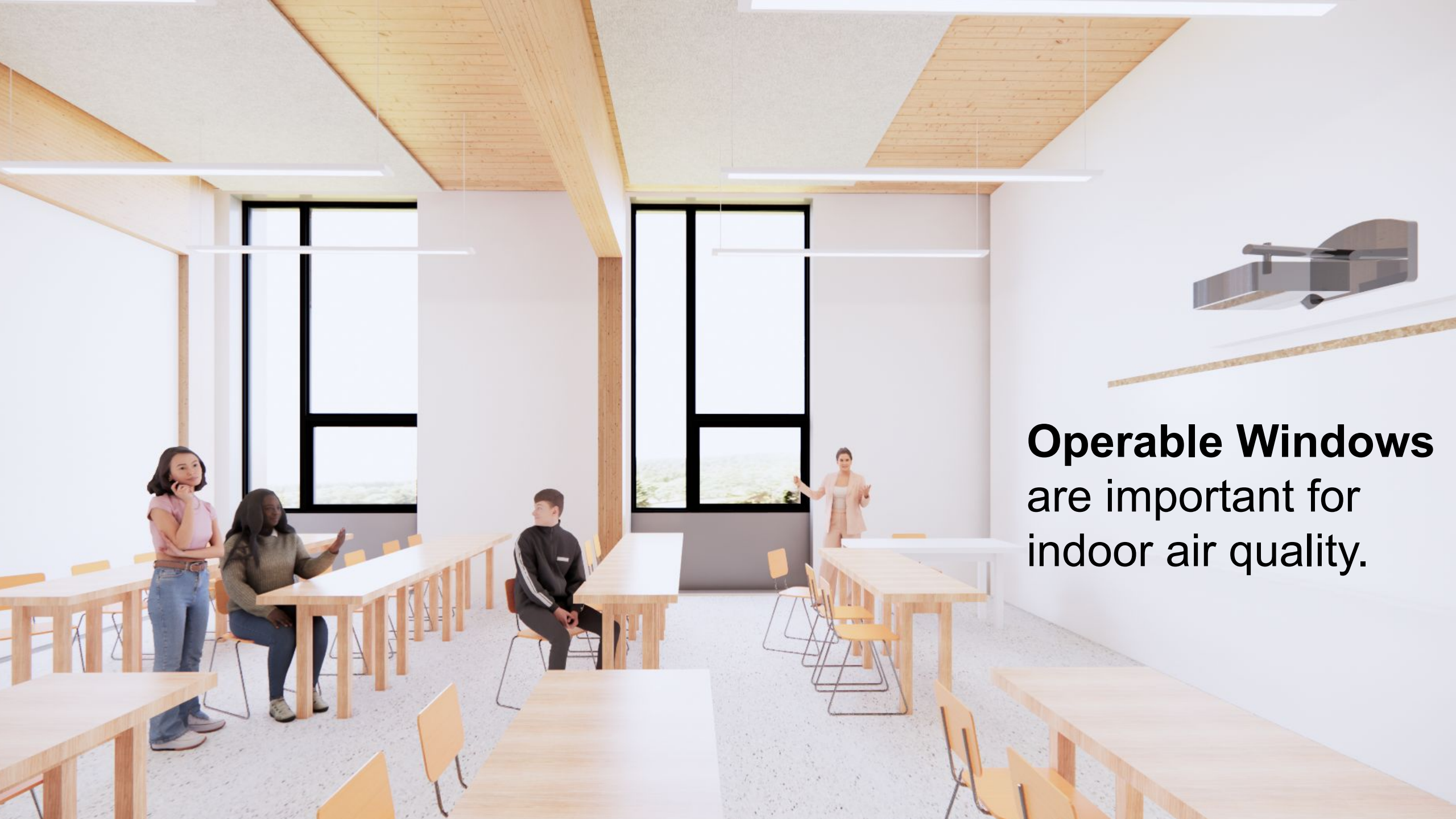
- Proposed CO2 Control Strategy:
  - Control to 800 ppm
  - 2-stage demand control ventilation strategy
    - Increase airflow to terminal unit
    - Open outside air damper at air handler
- Ventilation ASHRAE 62.1+30%
- Supply Air quantity:
  - 3.7-7.7 ACH
  - 0.8-1.3 CFM/sqft
- Supply Air filter rating: minimum MERV-13



ESTIMATED % OF HOURS ABOVE 800 PPM = 16% (71% DECREASE FROM BASE CASE)

ESTIMATED ENERGY:

— **EUI: 30.1**
1% increase



Operable Windows are important for indoor air quality.

PPS IAQ Design Summary for atypical conditions:

when outside air quality is compromised (Wildfires, high pollen, etc.)

—**Pandemic response mode**:
in-room HEPA filtration (portable)

—**Wildfire response mode**:
Reduce outside air to minimum values,
in-room HEPA filters



Purifier Inventory & Deployment History



Phase I: Initial Pandemic Response (2020)

- **Rapid Purifier Deployment:** Equipped every instructional space district-wide.
- **Strategic Allocation:**
  - **K-8 Spaces:** 4,500 Intellipure Ultrafine model #468 air purifier units
  - **High Schools:** 1,000 Medify 112 units (selected for high-capacity square footage)

Phase II: Tested Air Quality & Adapted Response (2021 - 2023)

- **Air Quality Report:** PBS Engineering (now APEX) and Ameresco conducted indoor air-quality tests.
  - Recommended 2 air purifiers per classroom at schools that lacked sufficient air exchanges (28 schools), which were distributed.
- **Procured Additional units:** April, 2023 - PPS Risk Management placed an order with OHA for 600 additional Medify 112 air purifiers, as well as 5 years of replacement filters for those purifiers and the Intellipure Ultrafine #468 units (OHA donation worth $5.8M).
  - PPS received approximately **3500x Medify 112 Pro units with 5 years of replacement filters for the Pro units.**
  - Additional units were communicated as available to any schools wanting secondary units in their classrooms.

Purifier Inventory & Deployment

Phase III: Iterate and Adapt (2024 - present)



- **Policies and Procedures:** With the Climate Crisis Response Policy (CCRP), PPS updated emergency response protocols and staff trainings to include heat and smoke in school closure and response events, seeking to miminize AQI hazards.
- **Communication**: Increased district-wide communication on best practices to maintain health in high smoke, heat, flu and pollen seasons.
- **Continued Maintenance and Distribution:** PPS custodians regularly vacuum air purifiers as needed, and PPS works with a contractor to replace filters and maintain units on an annual basis. Principals are encouraged to request additional air purifers for classrooms, as needed.
- **Investment in Systems:** PPS continues to update HVAC systems in priority schools with PCEF and bond investments.

Purifier Inventory & Deployment

Current State:

- To date over **6,000 air purifier units have been deployed.**
- PPS is currently storing about 2,500 Medify 112 Pro units and about 4 years of replacement filters in the warehouse, available upon request from building leaders.
- The maintenance team is deploying the Medify 112 Pro units where the original Intellipure and Medify 112 units no longer have replacement filters.
- Schools can, and do, regularly request additional air purifiers through facilities or the warehouse. These are deployed to the requesting school within a few days.
- Schools have the option to keep their existing units in addition to the new Pro units.

Challenges in Air Purifier Deployment

- **Deployment is highly variable:** The PPS warehouse regularly receives shipments of purifiers from schools that no longer want them.

- **Tracking**: Once purifiers are deployed to a school, they are sometimes moved to other locations without being reported centrally, so tracking of exact locations of deployment lacks precision over time.

- **Updated technology and surplus**: PPS would ideally procure filters for older Intellipure units that have used their replacement filters. However, this would be a large cost to the district when we already have an adequate supply of the newer Medify 112 Pro units and filters to deploy.

PCEF Grant Award and Projects

PPS has two large grant awards from the Portland Clean Energy Fund (PCEF)

- PCEF Strategic Program 16 Climate Friendly Public Schools (SP16), $19,930,833 (through 2028-29).
- PCEF's Collaborating for Climate Action (CCA), MESD is the fiscal agent, $25,568,240 for PPS (through 12/31/2030).

Collaborating for Climate Action (CCA) - January 1, 2026 - December 31, 2030

- Competitive grant, where MESD serves as the fiscal agent for all Portland school districts.
- Community Benefits Agreement with Coalition for the Communities of Color conduct community engagement
- Grant Budget includes:
  - Energy efficiency upgrades at two schools: (1) Upgrade HVAC: new air source heat pump, (2) Upgrade Water Heaters: electric heat pump, (3) Upgrade Windows.
  - Green schoolyard improvements and landscape maintenance for six schools
  - 1 FTE for Project Manager (Capital) + .5 FTE Green Schools Coordinator

PCEF Grant Award and Projects

PCEF SP16 - 5-Year Award 2024 - 2029, $19,930,833

- **Two funding components:** Student-led Climate Projects and Infrastructure Improvements
  - **$3,000,000 Student-led Climate Projects (Grades 6-12)**
    - **Purpose:** $10,000 per school (MS, HS, K-8) per year for 5 years for student-led projects, plus administrative funding
    - **Progress**:
      - 24-25: $317k total funds spent; 34/36 eligible schools participated; engaged over 4,300 students
      - 25-26: 100% of 37 eligible schools submitted project proposals, students plan to spend over $430k + administrative expenses.
        - Over 2,700 students involved in project development, expect to engage nearly 20,000 students



PCEF Grant Award and Projects

PCEF SP16 - 5-Year Award 2024 - 2029 - Infrastructure Improvements

- **PCEF SP16 - $16,930,833 Physical Improvement Projects**
  - **Purpose**: Increased comfort in the schools
    - Add cooling capability to address the increased frequency of warm/hot days during the school year
    - Install energy efficiency measures to save on utility costs
    - Decarbonize the system by switching from natural gas to electricity
    - HVAC improvements to start Summer 2027
  - November 2023: MOU with PAT commits $10 million of the PCEF Direct Allocation funds to address temperature mitigation at PCEF-qualified PPS buildings and to involve a PAT representative in reviewing relevant scope in the application (complete).

PCEF Grant Award and Projects

Infrastructure Improvements- Progress

SP16-Direct Allocation: Infrastructure	**Allocation**	**Description**
Phase I (2024 - 2025) Complete	$700,000	LED Lighting and weatherization upgrades (Complete)
Phase II (2025 - 2026)	$1,945,000	Design, abatement, hazardous materials testing in preparation for HVAC improvements
Phase III (2027 - 2029)	$14,285,833	Bulk of spend in future phases in construction and project implementation.

Questions?

